BIDSPHERE

SOFTWARE ENGINEERING (IT - 314)

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SYSTEM DESIGN

System Design Approach

Following the *Top-Down Design Approach* as outlined in the reference document:

1. **Begin with high-level system functionality, then decompose into cohesive subsystems.**
2. **Ensure high cohesion and low coupling through modular boundaries.**
3. **Design subsystems to be scalable, independent, and maintainable.**
4. **Emphasize performance, robustness, and usability.**
5. **Interfaces between subsystems follow clear contracts with consistent data formats.**

Design Goals

1. **Scalability: Support large concurrent auctions, bidders, and notifications.**
2. **Performance: Fast bidding updates (real-time), optimized database access.**
3. **Maintainability: Modular subsystems (auction, bidding, payments, admin).**
4. **Reliability: Ensure consistency during auto-bidding and auction closing.**
5. **User-friendly design: Easy bidding, clear payment flows, responsive UI.**
6. **Transparency & Trust: Verified sellers, ratings, admin moderation.**

Interface Design

External Interfaces

- **User Interface: Buyer & seller dashboard, Auction listing, Bidding UI.**
- **Admin Interface: Admin dashboard for verification & moderation.**
- **Payment Gateway: UPI QR + Link generator and transaction verifier.**
- **Email/Notification System: Winner emails, payment confirmations.**
- **Delivery Partner API: Tracking updates after successful payment.**
- **Storage System: User profile photos, item images.**

Inputs

- User credentials
- Auction details (title, images, reserve price, etc.)
- Bid amounts
- Payment confirmations (webhook payload)
- Reviews (ratings)
- Watchlist requests

Outputs

- Authentication responses
- Auction listings, bid history.
- Registration fee & payment links
- Winner notifications
- Delivery status
- Seller ratings

Key Relationships

- Users ↔ Platform: Create auctions, place bids, pay, track delivery.
- Admin ↔ Platform: Verify auctions, verify payments, cancel auctions, manage users.
- Buyers ↔ System: Register for auctions, bid, pay, add to watchlist.

Subsystem Decomposition

Subsystems are identified using partitioning (functional separation) and layering (logical grouping).

Subsystems and Their Interfaces

1. Authentication Subsystem

- Login, signup, forgot password
- Token generation & invalidation
- Role-based access (Admin / Buyer / Seller)
- Provides API for user registration and login and admin login

2. User Management Subsystem

- User profile management
- Profile photo upload
- Watchlist management

3. Auction Management Subsystem

- Auction creation, editing, cancellation
- Auction scheduling (start/end times)
- Auction verification (admin)
- Reserve price enforcement
- Automatic auction status transitions

4. Bidding & Auto-Bidding Subsystem

- Manual bidding
- Automatic bidding with max-limit
- Bid validation logic
- Highest bid tracking
- Real-time updates.

5. Payment Subsystem

- Registration fee calculation
- Winner payment (UPI QR + Link)
- Payment verification (admin)
- Payment status auditing

6. Delivery Subsystem

- Delivery creation after payment
- Courier API integration
- Delivery tracking
- Delivery completion

7. Notification Subsystem

- Winner emails
- Payment confirmation emails
- Web popup notifications
- Auction start/close alerts

8. Search & Filtering Subsystem

- Auction search
- Category filters
- Price-based sorting
- Live status filters

9. Review & Rating Subsystem

- Seller rating (1–5 stars)
- Rating summary aggregation

10. Admin Panel Subsystem

- Auction approval/rejection
- Payment verification
- Delivery verification
- User monitoring

Relationships Between Subsystems

- Data Coupling: Subsystems share auction, user, and payment records.
- Message/Event Coupling: Payment events trigger delivery & notifications.

Architectural Design

System Layers

1. Presentation Layer (Frontend)

- Buyer/Seller interface
- Admin dashboard
- Real-time bid updates
- Image uploads (auction items, profile photos)

2. Application Layer (Backend Services)

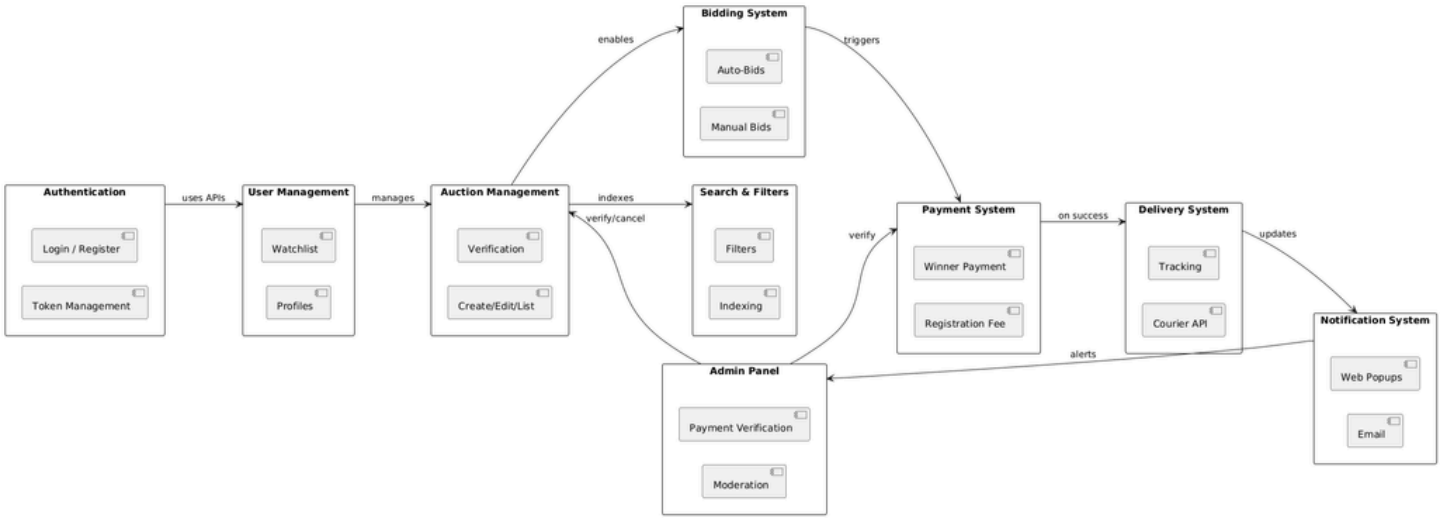
- Authentication Service
- User Management Service
- Auction Service

- **Bidding & AutoBid Service**
- **Payment Service**
- **Notification Service**
- **Delivery Service**
- **Watchlist Service**
- **Review Service**
- **Admin Service**

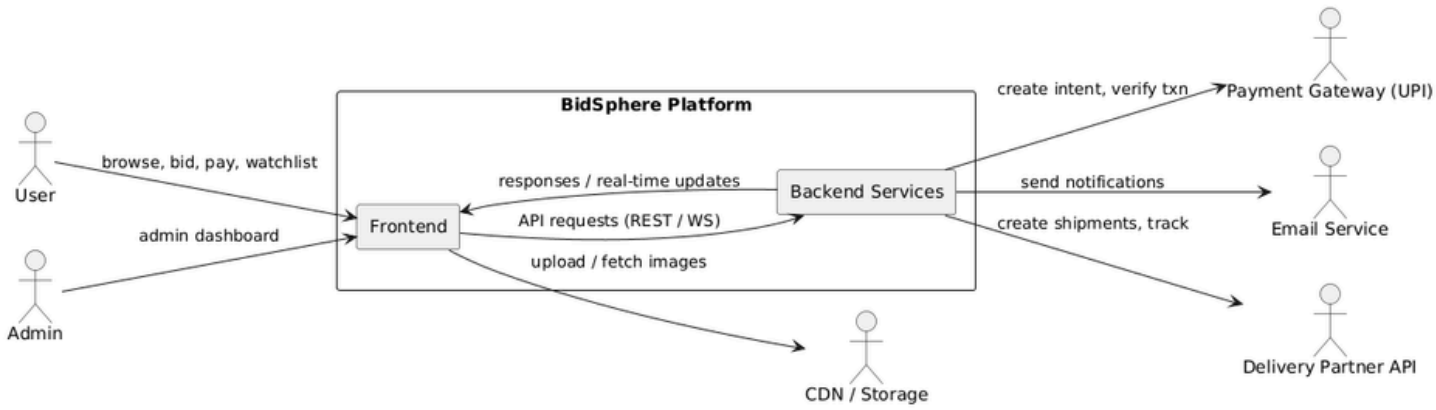
3. Data Layer

- **Primary Database: Mongodb atlas**
- **Cache: Redis for bid locks & auto-bid**
- **File Storage: CLOUDINARY for image store**

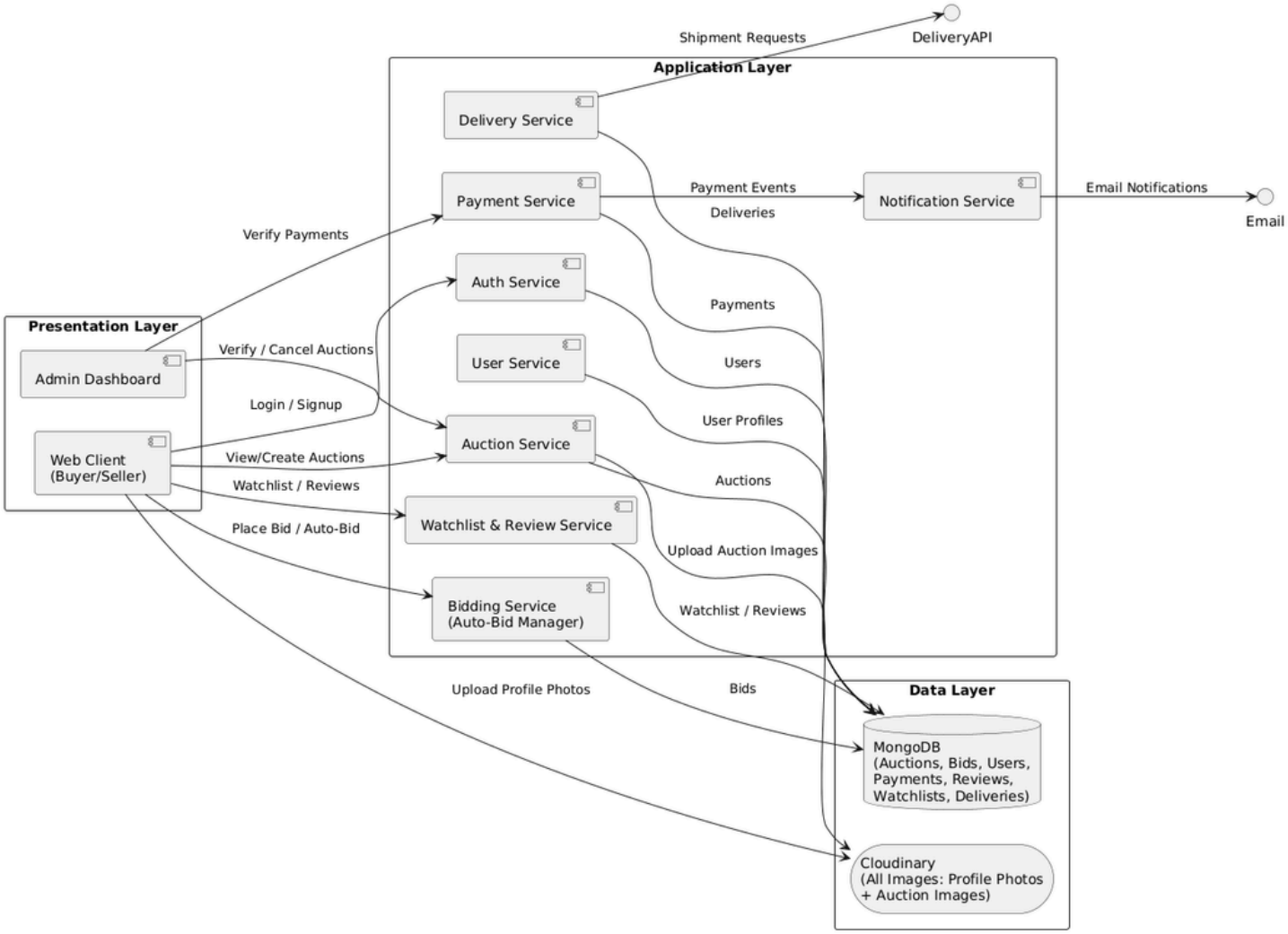
Module Relation Diagram:



System Context Diagram:



Subsystem Decomposition Diagram:


Shipment Requests
DeliveryAPI
Application Layer
Delivery Service
Payment Service
Payment Events
Deliveries
Notification Service
Email Notifications
Email
Verify Payments
Auth Service
Payments
Presentation Layer
Admin Dashboard
Verify / Cancel Auctions
User Service
Users
Login / Signup
User Profiles
Web Client
(Buyer/Seller)
View/Create Auctions
Auction Service
Auctions
Watchlist / Reviews
Watchlist & Review Service
Place Bid / Auto-Bid
Upload Auction Images
Bidding Service
(Auto-Bid Manager)
Watchlist / Reviews
Upload Profile Photos
Bids
Data Layer
MongoDB
(Auctions, Bids, Users,
Payments, Reviews,
Watchlists, Deliveries)
Cloudinary
(All Images: Profile Photos
+ Auction Images)